

4EB00 Q1 2026

Special Topic: The Jet Engine

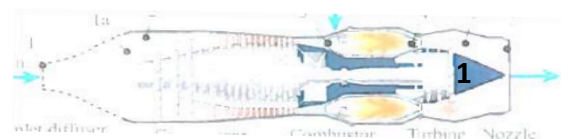
Groep **10**

	Naam	Studentnr
1		
2		
3		
4		

Assignment Conditions

Group 10 settings

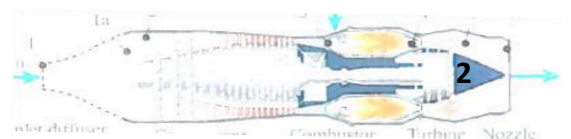
Name	Value	Unit
Fuel	H2	
Tamb	300	[K]
P3overP2	9	[-]
Pamb	100000	[Pa]
mfurate	0.58	[kg/s]
AF	204.42	[kg/kg]
v1	200	[m/s]



Assignment Cycle Analysis of the Jet Engine based on NASA tables

Applying conservation equations and the NASA tables fill out table 1. The code and approach should be explained in detail in the following pages. As an example a worked-out part of the procedure for the diffuser (page 5) is given. You should still add the code snippet and provide the actual line numbers.

State	1	2	3	4	5	6
P (kPa)	100.00	125.11	1125.98	1125.98	423.74	100.00
T (K)	300.00	319.78	591.58	1082.46	848.87	580.54
v (m/s)	200.00	0.00	0.00	0.00	0.00	778.83
Table 1: Thermodynamic state summary						



You do not have to fill this box. It will not be graded.

Code snippet diffusor [1-2] **the one in the sheets. Don't need yours here. Work starts at compressor stage**

Procedure (This just an Example. **Your work starts at the compressor stage.**)

Solved is first (**line nr**)

$$h_2(T_2) = h_1(T_1) + \frac{1}{2} v_1^2$$

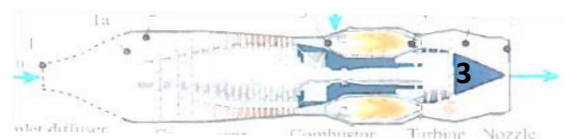
to determine T_2 the isentropic relation can be used to solve for the pressure yielding

$$\int_{T_{ref}}^{T_2} \frac{C_p(T)}{T} dT - R_g \ln\left(\frac{P_2}{P_{ref}}\right) = \int_{T_{ref}}^{T_1} \frac{C_p(T)}{T} dT - R_g \ln\left(\frac{P_1}{P_{ref}}\right).$$

Which is implemented in (**line nr**). Mixture properties determined using f.i. (**line nr**)

$$h(T) = \sum Y_i h_i(T)$$

The composition Y_i is taken equal to that of normal air.



Compressor

```
151 P3=P3overP2*P2;
152 s3sthermal=s2thermal+Rg*log (P3/P2);
153
154 T3s=invertProperty(sair_a,TR,s3sthermal,'Compressor T3s');
155 h3s=interpl (TR, hair_a, T3s);
156 S3s=s3sthermal-Rg*log (P3/Pref);
157
158 % eta_c=(h3s-h2)/(h3-h2)
159 h3=h2+(h3s-h2)/eta_c;
160 T3=invertProperty(hair_a,TR,h3,'Compressor T3');
161 s3thermal=interpl (TR, sair_a, T3);
162 S3=s3thermal-Rg*log (P3/Pref);
163
164 wcomp=h3-h2;           % Specific compressor work input [J/kg]
165 Wcomp=mair*wcomp;      % Compressor power input [W]
```

Code snippet Compressor [2–3], JetEngine_Group10.m lines 151–165

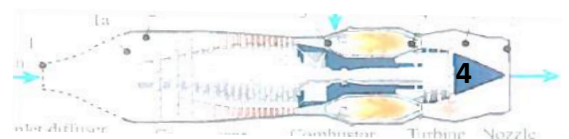
Procedure

The compressor is steady and adiabatic, with negligible changes in kinetic and potential energy. The prescribed compressor pressure ratio gives $P_3 = 9P_2$. For the isentropic reference state, $s_{3s} = s_2$, so $s_{T,3s} = s_{T,2} + R_{air} \ln(P_3/P_2)$. Line 154 inverts the NASA air-entropy curve to determine T_{3s} ; line 155 evaluates the corresponding enthalpy.

The efficiency definition is $\eta_c = (h_{3s} - h_2)/(h_3 - h_2)$. The lecturer confirmed $\eta_c = 1$, so $h_3 = h_{3s}$ and $T_3 = T_{3s}$. Lines 159–162 retain the general efficiency relation and evaluate the outlet state using the air mixture.

The steady-flow energy balance gives compressor input power $\dot{W}_c = \dot{m}_a(h_3 - h_2)$, where $\dot{m}_a = AF \times \dot{m}_f = 204.42 \times 0.58 = 118.5636$ kg/s. Lines 164–165 give a specific work input of 280.371 kJ/kg and compressor power of 33.241777 MW.

The outlet is $T_3 = 591.58$ K and $P_3 = 1125.98$ kPa. Both rise relative to state 2 as shaft work is supplied to the air.



Combustor (2 code snippets)

```

192 mprod=mair+mfurate;
193 mdot_in=mair*Yair+mfurate*Yfuel;
194 ndot_in=mdot_in./Mi;
195
196 nu=[-1 -0.5 0 1 0]; % Change per mole H2 burned
197 ndot_H2_burned=ndot_in(1);
198 ndot_prod=ndot_in+nu*ndot_H2_burned;
199 mdot_prod=ndot_prod.*Mi;
200
201 % Stoichiometric air-fuel ratio and equivalence ratio.
202 AF_st=(-nu(2)/Xair(2))*MAir/Mi(1);
203 phi=AF_st/AF;
204
205 %% Initial and final mixture composition
206 % Initial = air + fuel before combustion. Final = combustion products.
207 Yreac=mdot_in/sum(mdot_in);
208 Xreac=ndot_in/sum(ndot_in);
209 MReac=Xreac*Mi';
210 Rreac=Runiv/MReac;
211
212 Yprod=mdot_prod/sum(mdot_prod);
213 Xprod=ndot_prod/sum(ndot_prod);
214 MProd=Xprod*Mi';
215 Rprod=Runiv/MProd;

```

Composition before and after combustor [3–4], lines 192–215

```

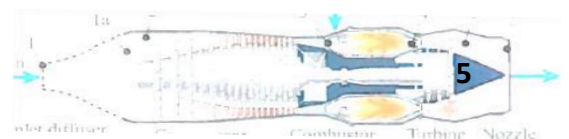
250 hprod_a=Yprod*hia';
251 sproda=Yprod*sia';
252
253 %% Combustor 3 -> 4
254 % Steady, adiabatic, no shaft work and negligible kinetic/potential energy.
255 % mair*h3+mfurate*hfuel=mprod*h4
256 % NASA enthalpies include formation enthalpy, so LHV is not added here.
257 hfuel=HNasa(Tfuel,SpS(1));
258 Hdot_in=mair*h3+mfurate*hfuel;
259 h4=Hdot_in/mprod;
260
261 T4=invertProperty(hprod_a,TR,h4,'Combustor T4');
262
263 % Combustion at constant pressure.
264 P4=P3;
265 s4thermal=interp1(TR,sproda,T4);
266 S4=s4thermal-Rprod*log(P4/Pref);

```

Thermodynamic computations [3–4], lines 250–266

Procedure

The species order is [H₂, O₂, CO₂, H₂O, N₂]. Mass flows are converted to molar flows using molecular masses M_i . For $\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$, the reaction vector is $[-1, -0.5, 0, 1, 0]$. Applying it to the incoming H₂ molar flow gives product molar flows; normalization gives X_{prod} and Y_{prod} . The pressure and thermal calculation are explained with Table 2 on the following page.



Combustor composition and procedure

AF = 204.42 (equivalence ratio = 0.16669)		
Mass fractions	Initial	Final
Fuel	0.004868	0.000000
O ₂	0.231784	0.193148
N ₂	0.763348	0.763348
CO ₂	0.000000	0.000000
H ₂ O	0.000000	0.043503
R _g [J/kg/K]	306.868	296.829
Table 2: Mixture composition before and after the combustor and the specific gas constants.		

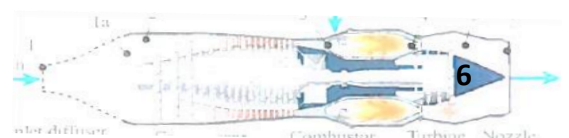
Initial denotes the combined, unburned air-plus-fuel mixture; final denotes the combustion products. It does not mean air alone at state 3. Fractions are rounded for display; the unrounded mass and mole fractions each sum to 1.

Procedure

The stoichiometric air-to-fuel ratio is $AF_{st} = 34.0741$ kg/kg, giving $\phi = AF_{st}/AF = 0.16669$. The mixture is lean, leaving excess oxygen. Nitrogen passes through unchanged, and there is no carbon in the fuel, so no CO₂ is formed. Total product flow is $\dot{m}_p = \dot{m}_a + \dot{m}_f = 119.1436$ kg/s. Mass and H/O/C/N atom balances close to numerical precision.

For the steady adiabatic combustor, shaft work and kinetic- and potential-energy changes are negligible. The correct reacting-flow balance is $\dot{m}_a h_3 + \dot{m}_f h_f = \dot{m}_p h_4$. The fuel enthalpy is evaluated with H_{Nasa} at 298.15 K (line 257); lines 258–261 determine h_4 and invert the product-enthalpy curve for T₄. NASA enthalpies include formation enthalpy, so adding an LHV term would double-count chemical energy.

The results are T₄ = 1082.46 K and P₄ = P₃ = 1125.98 kPa. Although $h_4 \approx 300.807$ kJ/kg is slightly below $h_3 \approx 302.278$ kJ/kg, the temperature rises because both composition and mass flow change. All subsequent calculations use $R_{prod} = 296.829$ J/(kg K) and the product property curves.



Turbine

```
327 v4=0;v5=0;
328 Wturb=Wcomp;
329
330 h5=h4-Wturb/mprod;
331 T5=invertProperty(hprod_a,TR,h5,'Turbine T5');
332
333 % eta_t=(h4-h5)/(h4-h5s)
334 h5s=h4-(h4-h5)/eta_t;
335 T5s=invertProperty(hprod_a,TR,h5s,'Turbine T5s');
336 s5sthermal=interp1(TR,sprod_a,T5s);
337
338 % S5s=S4 gives the turbine outlet pressure.
339 P5=P4*exp((s5sthermal-s4thermal)/Rprod);
340 S5s=s5sthermal-Rprod*log(P5/Pref);
341
342 s5thermal=interp1(TR,sprod_a,T5);
343 S5=s5thermal-Rprod*log(P5/Pref);
```

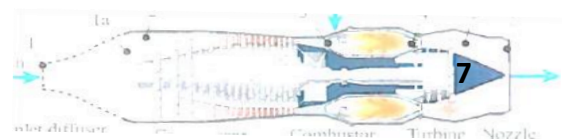
Code snippet Turbine [4–5], JetEngine_Group10.m lines 327–343

Procedure

The turbine drives only the compressor through a lossless shaft, so $\dot{W}_t = \dot{W}_c$. It processes $\dot{m}_p = 119.1436$ kg/s, whereas the compressor processes $\dot{m}_a = 118.5636$ kg/s. The balance is $\dot{m}_p(h_4 - h_5) = \dot{m}_a(h_3 - h_2)$, giving $h_5 = h_4 - \dot{W}_c/\dot{m}_p$ (line 330). Inverting the product-enthalpy curve yields T5.

The turbine efficiency is $\eta_t = (h_4 - h_5)/(h_4 - h_{5s})$. Lines 334–336 calculate the ideal reference state. With $\eta_t = 1$, $T_{5s} = T_5$. Its isentropic relation $s_{5s} = s_4$ gives $P_5 = P_4 \exp[(s_{T,5s} - s_{T,4})/R_{\text{prod}}]$, implemented at line 339. Product properties and the product gas constant must be used throughout.

The turbine outlet is $T_5 = 848.87$ K and $P_5 = 423.74$ kPa, with turbine power 33.241777 MW. Both temperature and pressure decrease, while the supplied power equals the compressor requirement.



Nozzle

```
371 P6=Pamb;
372 assert(P5>P6,'Nozzle: P5 must be above ambient pressure. ');
373
374 % Ideal nozzle state: S6s=S5.
375 s6sthermal=s5thermal+Rprod*log(P6/P5);
376 T6s=invertProperty(sprod_a,TR,s6sthermal,'Nozzle T6s');
377 h6s=interp1(TR,hprod_a,T6s);
378 S6s=s6sthermal-Rprod*log(P6/Pref);
379
380 % eta_n=(h5-h6)/(h5-h6s)
381 h6=h5-eta_n*(h5-h6s);
382 T6=invertProperty(hprod_a,TR,h6,'Nozzle T6');
383 s6thermal=interp1(TR,sprod_a,T6);
384 S6=s6thermal-Rprod*log(P6/Pref);
385
386 % h5+v5^2/2=h6+v6^2/2
387 v6s=sqrt(v5^2+2*(h5-h6s));
388 v6=sqrt(v5^2+2*(h5-h6));
```

Code snippet Nozzle [5–6], JetEngine_Group10.m lines 371–388

Procedure

The exit pressure is prescribed as ambient: $P_6 = 100$ kPa. For the isentropic reference, $s_{6s} = s_5$ gives $sT_{6s} = sT_5 + R_{\text{prod}} \ln(P_6/P_5)$. Lines 375–378 invert the product-entropy curve and evaluate h_{6s} . With $\eta_n = (h_5 - h_6)/(h_5 - h_{6s}) = 1$, $h_6 = h_{6s}$ and $T_6 = T_{6s}$.

The adiabatic nozzle has no shaft work. Its energy balance is $h_5 + v_5^2/2 = h_6 + v_6^2/2$. With v_5 neglected, $v_6 = \sqrt{2(h_5 - h_6)}$, using enthalpies in J/kg. The results are $T_6 = 580.54$ K and $v_6 = 778.83$ m/s. The enthalpy drop is approximately 303.286 kJ/kg; the negative $h_6 = -281.485$ kJ/kg is permitted by the NASA formation-enthalpy reference.

